FEDERICO COLOMBO

Computer Graphics Engineer | Technical Artist | XR Developer

Phone: +39-3513078191 | Email: colombofederico28@gmail.com | LinkedIn: linkedin.com/in/federico-colombo10 | Location: Turin, Italy

PROFESSIONAL PROFILE

Computer Graphics Engineer and Technical Artist specialized in real-time rendering, spatial computing (XR), and procedural tool development. Graduated with honors in Cinema and Media Engineering. Deep expertise in writing custom C++ rendering engines, optimizing GPU architectures (HLSL/GLSL), and leading multidisciplinary teams. Actively seeking to leverage hardcore engineering and project management skills to build scalable software architectures and high-fidelity interactive productions.

EDUCATION

M.Sc. Cinema and Media Engineering - Politecnico di Torino **2024 - 2026**

- **Expected Grade:** 110/110 Cum Laude (Graduating Dec 2026)

B.Sc. Cinema and Media Engineering - Politecnico di Torino **2021 - 2024**

- **Final Grade:** 100/110

WORK EXPERIENCE

Lemons in the Room & Politecnico di Torino | Firenze / Remote **03/2026 - Present**

Technical Artist & Producer (XR Master Thesis R&D)

- Architected a real-time bio-adaptive volumetric raymarching framework in HLSL for Meta Quest 3, rendering dynamic Signed Distance Fields (SDFs) driven by BLE biometric telemetry (**PulseEngine**).
- Engineered low-level GPU optimizations (Dynamic Bounding Volumes, adaptive ray stepping) locking **90 FPS on standalone XR hardware** (61.3% GPU time reduction, 2.91ms GPU time, 0B GC alloc).
- Authored **Morten Mikkelsen Surface Gradient** shading on procedural volumes (`∇_surf(h) = ∇h - (N · ∇h)N`) to eliminate triplanar UV seams, plus a sub-millisecond GPU Voxelizer (<0.8ms) for particle swarms.
- Architected a **Virtual DAW Mixer Console** (`DAWMixerWindow`) inside Unity with live OLED oscilloscopes and auto-packet sniffing for multi-sensor streams (BLE, OSC, WebSockets, Serial); mapped autonomic biofeedback along the **Bouba/Kiki spectrum** and modularized to official Unity Package (UPM) standards.

LEVEL UP LAB (PoliTo Student Team) | Torino, Italy **11/2024 - 07/2026**

Technical Artist, Gameplay Programmer & Project Manager

- Developed custom shaders in HLSL (retro CRT curvature, procedural water vertex displacement, cutaway systems) and core C# gameplay systems across team productions including *Line Operator* and *Spicy Mercante*.
- Modeled 3D assets in Blender and engineered full architectural refactorings (3D-to-2D engine transition, buoyancy physics, inventory systems).

- Coordinated multidisciplinary student pods of 6–7 developers as technical producer and project manager, enforcing Git LFS versioning and milestone sprints.

Politecnico di Torino | Turin, Italy **10/2024 - 01/2025**

Teaching Assistant (GUI)

- Tutored students in UI/UX principles and frontend programming during weekly laboratory sessions.
- Wrote technical documentation and created high-quality instructional content to improve comprehension.

RKH SRL | Turin, Italy **09/2023 - 06/2024**

Video Artist

- Operated professional cameras (RED, Blackmagic 4K/6K) and assisted DOP with complex lighting setups.
- Handled editing and post-production workflows using DaVinci Resolve for corporate content.

SKILLS

- **Languages:** C++, C#, Python, HLSL/GLSL
- **Engines:** Unity, Unreal Engine 5, Custom OpenGL Frameworks
- **Software:** Blender, DaVinci Resolve, Jira, Git
- **Management:** Project Management, Technical Direction, Agile Product Strategy

PERSONAL PROJECTS

- **Smart Cable Generator (Blender Tool):** Engineered a Python add-on for Blender to procedurally generate parametric hanging cables across multi-object modular assets with scale-independent catenary physics and automated mesh conversion for game engines.
- **Graphics Lab (Custom C++ Engine):** Engineered a custom OpenGL 3D rendering framework from scratch. Implemented Deferred Shading, Physically Based Rendering (PBR), and Silhouette Parallax Occlusion Mapping (SPOM) to study low-level GPU pipelines.
- **UE5 Tech Art Pipeline:** Developed procedural environments using PCG. Scripted a homing projectile system in Niagara and integrated MetaHumans with custom groom collision physics.
- **Character Rigging & Tooling:** Built complete custom FK/IK character rigs (Blender), painting skin weights and setting up constraint networks. Developed Python automation scripts to streamline 3D asset export pipelines into game engines.
- **Awaken Asylum & Nihil:** Developed an immersive VR educational experience (Unity) and a 2D Metroidvania prototype emphasizing complex transformation mechanics. Managed end-to-end technical production.